

AI Project Connect

Analytics dashboard — product & technical specification · v1.1 · 18 September 2026

Item	Detail
Product	AI Project Connect (Aptech)
Deliverable	A separate analytics dashboard at <code>/analytics</code>
For	The frontend developer building the dashboard
Back end	Built by the core team against the API contract in section 9 . Build the screens against it.
Status	Ready for build. The back end is live on the demo database (see section 18).

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1. Summary

AI Project Connect has three faces today:

- **Public website:** marketing pages, the courses catalogue and the idea application with its ₦2,000 commitment fee.
- **Client Portal:** clients sign in with a Project ID and a one-time code, then follow their build.
- **Engineering Panel** (/engineering): staff run projects, ideas, quotes, payments and the website content.

The panel already has a small **Reports** page, but it answers operational questions only. Nothing measures the website itself (visits, where people come from, what they click), and no screen shows the business end to end.

The **Analytics dashboard** is a separate, read-only product at /analytics . It answers one question per screen:

Screen	The question it answers
Overview	How is the business doing this period?
Traffic	How many people visit, from where, on what device?
Engagement	What do they click and use?
Funnels	Where do people drop out between visiting and paying?
Revenue	How much money came in, how much was refunded, and what is booked?
Projects	Are we delivering on time?
Sales pipeline	How do ideas turn into quotes and projects?
Clients	Who uses the portal, and how often?
Courses	Does "Learn the stack" turn into enrolments?
Team	How quickly does the team update clients and reply?
Operations	Are emails, SMS and payments working?
Realtime	What is happening right now?

Staff sign in with **the same email and password they use for the Engineering Panel**. The sign-in page is branded as Analytics.

2. Goals and non-goals

Goals

- **One trustworthy number per metric.** Every figure has a written definition ([section 6](#)), and the API computes it on the server. The browser never recomputes business numbers.
- **The whole funnel in one place,** from first visit to delivered project and course enrolment.
- **Money is labelled honestly.** Cash collected through the platform (commitment fees) is kept apart from booked value (accepted quotes, approved changes, support plans), because project invoices are paid outside the platform.
- **Everything can leave the dashboard:** CSV and Excel for every table, PDF for every screen, saved views, and a weekly emailed summary.
- **First-party and privacy-respecting tracking.** No Google Analytics and no third-party scripts, compliant with the Nigeria Data Protection Act 2023.

Non-goals (v1)

- **No editing of business data.** The dashboard is read-only; changes happen in the Engineering Panel.
 - **No per-person browsing history.** Analytics is aggregate: we never show "what did this visitor look at".
 - **No A/B testing, heatmaps or session replay.**
 - **No custom report builder.** Saved views cover the need for v1.
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3. Access and sign-in

3.1 Route

URL	Shows
/analytics	The sign-in page, or the Overview when already signed in
/analytics?view=<screen>	A screen, e.g. ?view=revenue . Keep the view in the query string — the site is a static export, so there are no dynamic routes.
/analytics?view=revenue&from=2026-08-01&to=2026-08-31&compare=previous	A shareable state. Every filter lives in the URL.

3.2 Who can see it

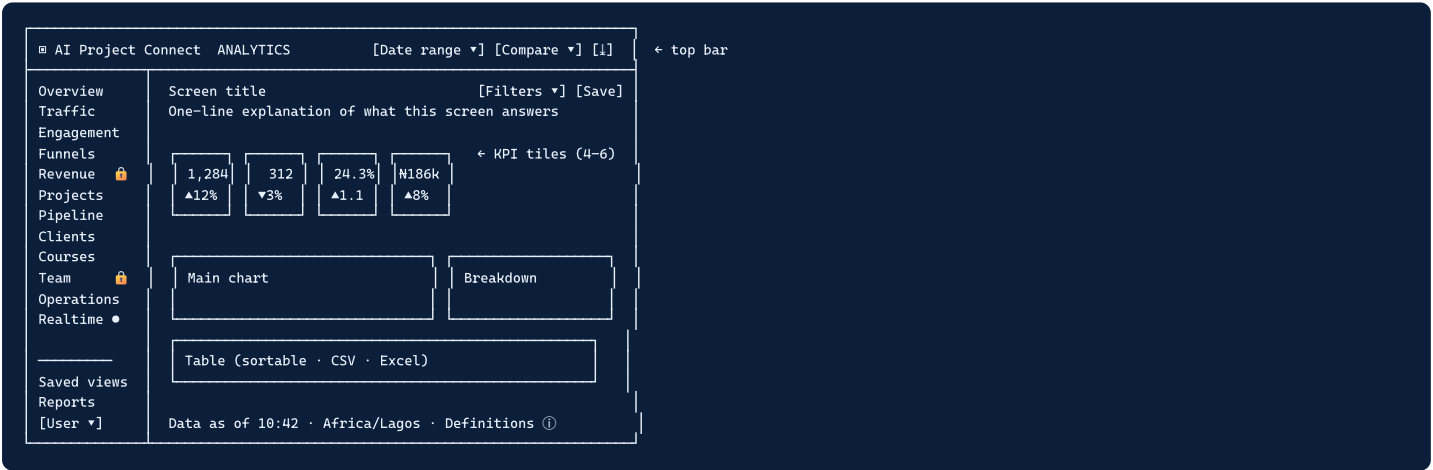
- **Admins** always have access.
- Any other active staff member has access when an admin ticks "**Can view analytics**" on their account (Engineering Panel → Users & roles). New API field: `canViewAnalytics` .
- **Revenue and Team** screens are admin-only, even for staff with analytics access. For everyone else the API returns `403` for those sections. Hide them from the navigation rather than showing an error.

3.3 The sign-in page

- **Same credentials and API** as the Engineering Panel: `POST /api/staff/auth/Login` with `{email, password}` . It is the same session cookie, so signing in to one signs you in to both.
- **Branding:** the product mark plus an "**Analytics**" badge. Title "Sign in to Analytics", subtitle "Use your AI Project Connect staff account."
- "**Forgot password?**" uses the existing flow (`/api/staff/auth/forgot-password`).
- **After sign-in**, call `GET /api/analytics/me` (9.2). If `canViewAnalytics` is `false` , show "Your account doesn't have access to Analytics. Ask an admin to turn it on." and a **Sign out** button. Never show a half-loaded dashboard.
- **Session expiry:** any `401` from an analytics endpoint returns to the sign-in page and keeps the URL, so the user lands back on the same view after signing in.
- **No self-sign-up.** Accounts are created by admins in the Engineering Panel.

4. Screens

Every screen follows the same frame:



- **Layout:** the sidebar collapses to a top menu below 1024px. Tiles wrap 4 → 2 → 1 column; charts go full width on mobile. No horizontal page scroll at any width; wide tables scroll inside their card.
-  marks **admin-only** screens (see 3.2).
- **Footer on every screen:** "Data as of HH:mm", the timezone, and a **Definitions** link that opens the metric dictionary entries for that screen in a side drawer.

The sections below list the exact content of each screen. Metric names match [section 6](#); endpoints match [section 9](#).

4.1 Overview — ?view=overview

The one-page business summary. Load it first; it has to feel instant.

Block	Content
KPI tiles	Visitors · Ideas submitted · Fee revenue (net) · Booked contract value · Active projects · On-time delivery rate
Trend	Visitors and ideas submitted per day. Show them as two small charts stacked, never on one dual-axis chart .
Funnel strip	Visitors → Idea form opened → Fee paid → Submitted → Quoted → Accepted, with the conversion % between steps
Attention list	Overdue projects, transfers waiting for confirmation, refunds due, stale projects (no client update in 5+ days). Each row links to the Engineering Panel.
Top sources	The top 5 traffic sources, with visitors and the application conversion rate

Endpoint: GET /api/analytics/overview .

4.2 Traffic — ?view=traffic

Block	Content
KPI tiles	Visitors · Sessions · Page views · Pages per session · Avg. session duration · Bounce rate
Main chart	A time series of the selected metric (a toggle picks the metric), with the comparison period as a thin dashed line
Breakdowns (tabs)	Source · Medium · Campaign (UTM) · Referrer · Landing page · Page · Device · Browser · OS · Country · State (Nigeria)
When people visit	A 7 × 24 heatmap: day of week × hour, Africa/Lagos time
New vs returning	Share of visitors seen before

Endpoints: GET /api/analytics/traffic , /traffic/timeseries , /traffic/breakdown .

4.3 Engagement — ?view=engagement

Block	Content
KPI tiles	CTA clicks · Tracker searches · Idea forms opened · Course clicks · Outbound clicks
Top interactions	Table of event name × target (e.g. cta_click · hero_submit_idea) with clicks, unique visitors, and click-through rate relative to that page's views
Tracker searches	Searches by kind (Project ID or idea ref) and result (found / not found / rate-limited)
Scroll depth	The share of page views reaching 25 / 50 / 75 / 100% on the home page
Downloads	Report PDFs, proposals and shared files downloaded (counts only)

Endpoint: GET /api/analytics/engagement .

4.4 Funnels — ?view=funnels&funnel=<id>

Four funnels, chosen with a segmented control. Each shows the steps with counts, step-to-step conversion, overall conversion, and the median time between steps. Every step count is **unique people** (visitors or applications), not events.

Funnel id	Steps
application	Visited → Opened idea form → Draft saved → Reached payment → Fee paid (or transfer reported) → Submitted
sales	Submitted → Reviewed → Quote sent → Quote accepted → Project started → Delivered
portal	Tracker search → Code requested → Code verified (signed in) → Returned within 30 days
courses	Course viewed → Course clicked → Enquiry / request → Contacted → Enrolled

Below the chart, a table breaks the same funnel down by **source**, **device** or **category** (a toggle). Endpoint: `GET /api/analytics/funnels/{id}`.

4.5 Revenue — ?view=revenue admin only

Important wording. The platform collects only the commitment fee. Project invoices, course fees and support plans are paid outside it, so this screen splits **cash received** from **booked value**. Never add the two together into one "revenue" total.

Block	Content
Cash tiles	Fees collected (gross) · Refunded · Net fee revenue · Transfers awaiting confirmation (count and £)
Booked tiles	Contract value won · Approved change requests · Support plan MRR · Estimated course revenue
Cash over time	Net fee revenue per day, week or month, as columns
By payment method	Paystack · Bank transfer · Paid at centre — amount and share
Refunds	Count, amount, median days from decline to refund
Quote value	Quoted vs accepted value over time, plus the quote acceptance rate
Ledger table	Every payment in range: date, reference, receipt, idea, method, amount, status, refund status. Exportable.

Endpoint: `GET /api/analytics/revenue`.

4.6 Projects — ?view=projects

Block	Content
KPI tiles	Active projects · Delivered (in range) · On-time delivery rate · Median cycle time · Overdue now · Avg. client rating
Projects by stage	Horizontal bars: Approved, Design, Development, Testing, Deployment, On hold
Time in stage	The median days spent in each stage, from stage history
Delivery vs target	Per delivered project: days early or late (a diverging bar)
Update frequency	Client-visible updates per project in the last 30 days; stale projects highlighted
Table	Project, lead, stage, progress, target date, days overdue, last client update, rating

Endpoint: `GET /api/analytics/projects`.

4.7 Sales pipeline — ?view=pipeline

Block	Content
KPI tiles	Ideas submitted · Drafts started · Draft completion rate · Median time to first quote · Quote acceptance rate · Walk-in share
Ideas over time	Submitted per week, split online vs walk-in
By category / platform / budget / state	Bars, with a toggle for the dimension
Ageing	Submitted ideas not yet quoted, bucketed 0-2, 3-7, 8-14 and 15+ days
Table	Ref, title, category, source, submitted, fee status, stage in the pipeline, days waiting

Endpoint: `GET /api/analytics/pipeline`.

4.8 Clients — ?view=clients

Block	Content
KPI tiles	New clients · Active clients (signed in during the range) · Portal sign-ins · Returning-client rate · Messages from clients · Median team reply time
Sign-ins over time	Per day
Engagement	Milestone approvals, change requests raised, files uploaded, course requests, ratings given
Opt-outs	Share opted out of weekly emails and of course suggestions
By location	Clients by state and country

Endpoint: `GET /api/analytics/clients`.

4.9 Courses — ?view=courses

Block	Content
KPI tiles	Course views · Clicks · Enquiries · Enrolled · View-to-enrol rate · Estimated course revenue
Funnel by course	Table: course, views, clicks, enquiries, contacted, enrolled, conversion
Lead sources	Portal · Website · Invite
Counsellor performance	Leads handled, median time to first contact, enrolment rate

Endpoint: GET /api/analytics/courses .

4.10 Team — ?view=team admin only

Block	Content
KPI tiles	Updates posted · Median approval time (engineer update → published) · Median client-reply time · Quotes sent · Payments confirmed
Per person	Table: name, role, projects, updates, internal notes, replies, median reply time, approvals given
Workload	Active projects per lead and engineer

Use this to spot overload and slow spots, not to rank people. Show the caption "Use these numbers to spot overload and slow spots, not to rank people" at the top of the screen.

Endpoint: GET /api/analytics/team .

4.11 Operations — ?view=operations

Block	Content
Messaging	Emails and SMS by status (sent, failed, queued) per day; failure rate; top failure reasons
Payments	Paystack attempts, success rate, abandoned checkouts; median time to confirm a transfer
Security	Staff sign-ins, failed sign-ins, password resets, one-time codes requested vs verified, rate-limit hits
API health	Error responses (4xx/5xx) per day, the slowest endpoints (p95 ms)

Endpoint: GET /api/analytics/operations .

4.12 Realtime — ?view=realtime

Refreshes every 15 seconds while the tab is visible, and pauses when it is hidden.

Block	Content
Now	Active visitors (last 5 minutes), large
Last 30 minutes	Page views per minute as columns
Right now	Top pages, top sources, devices
Live feed	The last 50 events, anonymised: time, event, page, source, device. No visitor ids and no IPs.

Endpoint: GET /api/analytics/realtime .

4.13 Saved views and Reports

See [section 10](#).

5. Global controls

Control	Behaviour
Date range	Presets: Today, Yesterday, Last 7 days, Last 30 days (default), Last 90 days, This month, Last month, This year, Custom. Always inclusive, always in Africa/Lagos time. Sent as <code>from</code> and <code>to</code> (<code>YYYY-MM-DD</code>).
Compare	None · Previous period (default) · Same period last year. Tiles show the delta (▲/▼ with %); charts show the comparison as a thin dashed line.
Interval	Auto (hour when the range ≤ 2 days, day ≤ 90 days, week ≤ 1 year, month beyond), with a manual override.
Filters	Source, medium, campaign, device, country, state, idea category, payment method. Show a filter only on the screens where it applies (the API lists them per screen in <code>meta.filters</code>). Active filters appear as removable chips under the title.
Currency	NGN. Format as <code>₦1,250,000</code> (no decimals) or <code>₦1.25M</code> in compact tiles.
Numbers	Thousands separators. Percentages to one decimal place. Durations as <code>3d 4h</code> , <code>2h 10m</code> or <code>45s</code> .
Share	Every state lives in the URL, so copying it shares exactly what you see.

A delta is **good** or **bad** according to each metric's `goodDirection` in the API meta. For example, bounce rate going up is bad. Colour the delta with that, never with the arrow direction.

6. Metric dictionary

All counts are for the selected range in Africa/Lagos time unless the metric says otherwise.

- **Source** says where the number comes from. `events` means browser tracking (section 7); anything else is a table in the database. Numbers taken from the database are authoritative.
- **Traffic numbers start from the day tracking goes live.** Show "No data before " rather than zeros for earlier dates. The API returns `trackingSince` in the response meta.

6.1 Traffic

Metric	Definition	Source	Good ↑/↓
Visitors	Distinct <code>visitor_id</code> with at least one event	events	↑
New visitors	Visitors whose first-ever event falls in the range	events	↑
Returning visitors	Visitors in range whose first event is before the range	events	↑
Sessions	Distinct <code>session_id</code> . A session ends after 30 minutes without activity, or at midnight Lagos time.	events	↑
Page views	Count of <code>page_view</code> events	events	↑
Pages per session	Page views ÷ sessions	events	↑
Avg. session duration	Mean of (last event – first event) per session. Single-event sessions count as 0.	events	↑
Bounce rate	Sessions with exactly one <code>page_view</code> and no other interaction event, ÷ sessions	events	↓
Source	The UTM source if present, otherwise the referrer's domain grouped by the source rules (7.5), otherwise <code>direct</code>	events	—

6.2 Engagement

Metric	Definition	Source	Good
CTA clicks	<code>cta_click</code> events	events	↑
CTA click-through rate	Unique visitors who clicked a CTA ÷ unique visitors who viewed the page it sits on	events	↑
Tracker searches	<code>tracker_search</code> events, split by <code>kind</code> and <code>result</code>	events	—
Not-found rate	Tracker searches with <code>result=not_found</code> ÷ all tracker searches	events	↓
Scroll depth	Share of home-page views reaching each threshold	events	↑

6.3 Application funnel and pipeline

Metric	Definition	Source	Good
Idea forms opened	Unique visitors with <code>idea_form</code> <code>step=opened</code>	events	↑
Drafts started	Ideas created with status DRAFT in range (<code>ideas.created_at</code>)	ideas	↑
Draft completion rate	Drafts created in range that were later submitted ÷ drafts created in range	ideas	↑
Ideas submitted	<code>ideas.submitted_at</code> in range (never DRAFT)	ideas	↑
Walk-in share	Submitted with <code>source=walk_in</code> ÷ submitted	ideas	—
Time to first quote	Median of (<code>quotes.created_at</code> of the first quote – <code>ideas.submitted_at</code>)	quotes, ideas	↓
Quote acceptance rate	Quotes accepted ÷ quotes that are no longer open (accepted + declined + withdrawn + expired)	quotes	↑
Idea → project conversion	Submitted ideas with a <code>project_id</code> ÷ submitted ideas (cohort by submission date)	ideas	↑

6.4 Revenue — cash vs booked

Metric	Definition	Source	Kind
Fees collected (gross)	Sum of <code>amount_kobo</code> ÷ 100 where <code>status=PAID</code> and <code>paid_at</code> in range	idea_payments	Cash
Refunded	Sum where <code>refund_status=REFUNDED</code> and <code>refunded_at</code> in range	idea_payments	Cash
Net fee revenue	Fees collected – Refunded, each by its own date	idea_payments	Cash
Refunds due	Count and sum where <code>refund_status</code> IN (<code>PENDING</code> , <code>PROCESSING</code>) right now (not range-bound)	idea_payments	Liability
Awaiting confirmation	Count and sum where <code>status=AWAITING_CONFIRMATION</code> right now	idea_payments	Pending cash
Payment success rate	<code>PAID</code> ÷ (<code>PAID</code> + <code>FAILED</code> + abandoned <code>PENDING</code> older than 1 hour), Paystack only	idea_payments	↑
Contract value won	Sum of <code>quotes.amount</code> where <code>status=accepted</code> and <code>responded_at</code> in range	quotes	Booked
Quoted value	Sum of <code>quotes.amount</code> where <code>created_at</code> in range	quotes	Pipeline
Approved change requests	Sum of <code>impact_cost</code> where status <code>APPROVED</code> or <code>COMPLETED</code> and <code>decided_at</code> in range	change_requests	Booked
Support plan MRR	Sum of the monthly price of the current <code>support_plan</code> for delivered projects, with prices from site content <code>supportPlans</code> . A snapshot "as of today".	projects, site_content	Booked (recurring)

Metric	Definition	Source	Kind
Estimated course revenue	Sum of the discounted course price for leads that became ENROLLED in range	leads, courses	Estimate

Every money metric in the API carries a `kind` field: `cash`, `booked`, `estimate`, `liability` or `pending`. Show that kind as a small label on the tile.

6.5 Projects and delivery

Metric	Definition	Source	Good
Active projects	Stage not DELIVERED, as of the range end	projects	—
Delivered	<code>delivered_at</code> in range	projects	↑
On-time delivery rate	Delivered in range with <code>delivered_at ≤ target_date</code> ÷ delivered in range	projects	↑
Cycle time	Median of (<code>delivered_at</code> – <code>start_date</code>) in days, for projects delivered in range	projects	↓
Time in stage	Median days between entering and leaving each stage	project_stage_history (new, 8.2)	↓
Overdue now	Active projects with <code>target_date < today</code>	projects	↓
Stale projects	Active projects with no client-visible published update for 5+ days	updates	↓
Updates per project	Client-visible published updates per active project over the last 30 days	updates	↑
Avg. client rating	Mean <code>rating_stars</code> of projects rated in range	projects	↑

6.6 Clients

Metric	Definition	Source	Good
New clients	<code>clients.created_at</code> in range	clients	↑
Portal sign-ins	One-time codes verified (<code>otp_codes.consumed_at</code>) in range	otp_codes	↑
Active clients	Distinct clients with at least one sign-in in range	otp_codes → projects	↑
Returning-client rate	Clients active in range who were also active in the previous 30 days ÷ active clients	otp_codes	↑
Messages from clients	<code>messages.sender=client</code> in range	messages	—
Median team reply time	Per client message: time until the next team message on the same project. The median over messages in range; unanswered messages are excluded and counted separately.	messages	↓

6.7 Courses

Metric	Definition	Source	Good
Course views / clicks	<code>course_events</code> with event <code>view / click</code>	course_events	↑
Enquiries	Leads created in range (any source)	leads	↑
Enrolled	Leads whose status became ENROLLED in range	leads	↑
View-to-enrol rate	Enrolled ÷ course views (same range)	both	↑
Time to first contact	Median of (first status change from NEW – <code>created_at</code>)	leads + activity_log	↓

6.8 Team

Metric	Definition	Source	Good
Updates posted	Updates by <code>author_id</code> , client-visible and internal counted separately	updates	—
Approval time	Median of (<code>published_at</code> – <code>created_at</code>) for updates that were pending	updates	↓
Replies	Team messages by <code>user_id</code>	messages	—
Workload	Active projects where the person is the lead or a member	projects, project_members	—

6.9 Operations

Metric	Definition	Source	Good
Messages sent / failed	<code>notifications</code> by status and channel, <code>created_at</code> in range	notifications	failed ↓
Delivery failure rate	failed ÷ (sent + failed)	notifications	↓
Codes requested vs verified	<code>otp_codes</code> created vs consumed	otp_codes	—
Rate-limit hits	Requests refused with 429	api_request_log (new)	—
API errors	Responses ≥ 500, and 4xx excluding 401/404	api_request_log (new)	↓
p95 latency	95th percentile response time per endpoint group	api_request_log (new)	↓

7. Tracking: what the website records

Tracking is **first-party** and **batched**. A small script in the existing site (built by the core team) collects events and sends them to `POST /api/track`. The analytics dashboard only reads.

7.1 Identifiers

Id	What	Where	Lifetime
<code>visitor_id</code>	A random UUID v4. It has no link to a person.	First-party cookie <code>apc_vid</code> (and <code>localStorage</code> as a fallback)	13 months, renewed on visit
<code>session_id</code>	A random UUID v4	<code>sessionStorage</code>	Ends after 30 minutes idle or at midnight Lagos time

Rules:

- **Never** put a name, email, phone, Project ID, idea reference or IP into an event.
- The server links a signed-in client or staff member to a visit only in aggregate, e.g. "sessions that signed in to the portal". It never exposes that link in any response.

7.2 Page URL hygiene

Before sending, the tracker **removes** these query parameters from `path` and `referrer`: `resume`, `token`, `reset`, `reference`, `ref`, `code`, `email`. They are private links. It keeps `utm_*`, `view`, `funnel` and `payment`.

7.3 Event catalogue

Event	When	Properties
<code>page_view</code>	Every route change, including <code>?view=</code> changes	<code>path</code> , <code>title</code>
<code>cta_click</code>	A tracked call-to-action is clicked	<code>id</code> (see 7.4), <code>path</code>
<code>tracker_search</code>	The home-page tracker is submitted	<code>kind</code> : <code>project/idea</code> · <code>result</code> : <code>found/not_found/rate_limited/error</code>
<code>portal_signin</code>	Client sign-in steps	<code>step</code> : <code>code_requested/verified/failed</code>
<code>idea_form</code>	Application steps	<code>step</code> : <code>opened/about_you/idea/budget/payment/draft_saved/resume_link_requested/submitted</code> · <code>variant</code> : <code>modal/page</code>
<code>payment</code>	Wallet actions	<code>method</code> : <code>paystack/manual</code> · <code>step</code> : <code>started/returned_success/returned_failed/transfer_reported</code>
<code>course_view</code> / <code>course_click</code>	Course cards	<code>courseId</code>
<code>quote</code>	Quote page	<code>step</code> : <code>viewed/accepted/declined</code>
<code>download</code>	A file download	<code>kind</code> : <code>report/proposal/file</code>
<code>outbound_click</code>	A link to another site	<code>host</code> (not the full URL)
<code>scroll_depth</code>	Home page thresholds	<code>depth</code> : <code>25/50/75/100</code>
<code>client_error</code>	An unexpected UI error	<code>code</code> (no message text)

Business facts (payments, submissions, quotes, projects) are **not** tracked from the browser. They are read from the database, which is the source of truth.

7.4 CTA ids

`nav_submit_idea`, `hero_track`, `flier_cta`, `course_enrol`, `course_info`, `footer_cta`, `announcement_link`, `portal_learn_this`, `portal_invite`, `apply_pay_paystack`, `apply_pay_transfer`, `apply_save_later`.

To add one: give the element `data-track=<id>` and the tracker picks it up.

Each event also carries a `name` (the CTA id, the step, or the course id) so the API can group without reading `props`. `idea_form` steps are sent in this order: `opened` → `about_you` → `idea` → `budget` → `payment` → `draft_saved` / `resume_link_requested` → `submitted`.

7.5 Source rules

1. If `utm_source` is present, use it (lower-cased). The medium comes from `utm_medium`, else `referral`.
2. Otherwise group the referrer domain:
 - **search**: `google`, `bing`, `duckduckgo`, `yahoo`, `yandex`
 - **social**: `facebook`, `instagram`, `x/twitter`, `linkedin`, `tiktok`, `whatsapp`, `youtube`
 - **email**: `mail.google`, `outlook`
 - Anything else keeps its domain as a **referral**.
3. No referrer (or our own domain) means `direct`.

7.6 Transport and filtering

- **Batching**: events are sent in batches of up to 20, and immediately on `pagehide` using `navigator.sendBeacon`.
- **Server-side filtering**:
 - Payload cap: 32 KB.
 - Rate limit: 120 events per minute per IP.

- Bots are dropped by user-agent list and headless markers.
 - Staff sessions are flagged `internal=1` and excluded by default. There is a toggle "Include staff traffic".
 - **Opt-out:** tracking is off when Do Not Track or Global Privacy Control is set, or when the visitor turns it off in the cookie notice.
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8. Data model

8.1 Existing tables the dashboard reads

projects , project_members , clients , ideas , idea_payments , quotes , change_requests , updates , messages , milestones , leads , courses , course_events , otp_codes , notifications , activity_log , users , site_content .

8.2 New tables (built by the core team)

```
-- One row per tracked browser event (raw, 13-month retention)
analytics_events (
  id          BIGINT UNSIGNED PK,
  occurred_at DATETIME(3) NOT NULL,      -- client time, clamped to server time ± 10 min
  received_at DATETIME(3) NOT NULL,
  visitor_id  CHAR(36) NOT NULL,
  session_id  CHAR(36) NOT NULL,
  event       VARCHAR(40) NOT NULL,      -- page_view, cta_click, ...
  name        VARCHAR(80) NULL,          -- cta id, funnel step, course id ...
  path        VARCHAR(255) NULL,         -- cleaned (7.2)
  props       JSON NULL,                -- whitelisted keys only
  source      VARCHAR(80) NULL, medium VARCHAR(40) NULL, campaign VARCHAR(120) NULL,
  referrer_host VARCHAR(120) NULL,
  device      ENUM('desktop','mobile','tablet') NULL,
  browser     VARCHAR(40) NULL, os VARCHAR(40) NULL,
  country     CHAR(2) NULL, state VARCHAR(80) NULL, -- from IP, IP not stored
  internal    TINYINT(1) NOT NULL DEFAULT 0, -- staff traffic
  KEY (occurred_at), KEY (event, occurred_at), KEY (session_id), KEY (visitor_id)
)

-- One row per session, updated as events arrive
analytics_sessions (
  session_id CHAR(36) PK, visitor_id CHAR(36), started_at, ended_at, page_views INT,
  events INT, landing_path, exit_path, source, medium, campaign, referrer_host,
  device, browser, os, country, state, is_bounce TINYINT(1), internal TINYINT(1),
  signed_in_as ENUM('none','client','staff')
)

-- First-seen date per visitor (new vs returning)
analytics_visitors (visitor_id CHAR(36) PK, first_seen DATETIME, last_seen DATETIME)

-- Daily rollups so long ranges stay fast (kept forever)
analytics_daily (day DATE, metric VARCHAR(40), dimension VARCHAR(40), value_key VARCHAR(160),
  value BIGINT, PRIMARY KEY (day, metric, dimension, value_key))

-- Stage history, for time-in-stage (back-filled from existing stage updates)
project_stage_history (id, project_id, from_stage, to_stage, changed_at, changed_by)

-- Lightweight request log, for the Operations screen (30-day retention)
api_request_log (id, at DATETIME(3), method, route, status SMALLINT, ms INT, internal TINYINT(1))

-- Dashboard features
analytics_saved_views (id, user_id, name, query JSON, shared TINYINT(1), created_at)
analytics_schedules (id, user_id, name, view, query JSON, frequency ENUM('daily','weekly','monthly'),
  recipients JSON, format ENUM('pdf','csv','xlsx'), last_sent_at, active TINYINT(1))

-- Access flag
ALTER TABLE users ADD can_view_analytics TINYINT(1) NOT NULL DEFAULT 0;
```

8.3 Jobs (cron)

Job	Schedule	Does
bin/analytics-rollup.php	Every night at 01:00 Lagos	Rolls up yesterday into analytics_daily , closes sessions, and re-rolls the last 3 days to catch late events
bin/analytics-reports.php	Every hour	Sends due scheduled reports
bin/analytics-prune.php	Every night	Deletes raw events older than 13 months and request logs older than 30 days

Query strategy. Today and yesterday are read from raw events; older days come from the rollups. For a mixed range the API stitches both together. The frontend never needs to care about this.

9. API contract

- **Base path:** /api/analytics .
- **Transport:** JSON over the same session cookie as the Engineering Panel.
- **Requests:** send credentials: "include" on every request. Every non-GET request also needs the header X-Requested-With: XMLHttpRequest .
- **Errors:** { "error": "message", "errors": { "field": "message" } }
 - 401 : signed out.
 - 403 : no analytics access, or an admin-only section.
 - 422 : bad parameters.
 - 429 : too many requests.

9.1 Common query parameters

Param	Values	Default
from , to	YYYY-MM-DD (Africa/Lagos, inclusive)	last 30 days
compare	none , previous , year	previous
interval	auto , hour , day , week , month	auto
source , medium , campaign , device , country , state , category , method	a filter value	—
includeInternal	1 to include staff traffic	0

Maximum range: 2 years. Hour interval only for ranges ≤ 7 days.

9.2 Session

GET /api/analytics/me

```
{
  "user": { "id": 1, "name": "Aptech Dev Team", "role": "admin" },
  "canViewAnalytics": true,
  "sections": ["overview","traffic","engagement","funnels","revenue","projects","pipeline","clients","courses","team","operations","realtime"],
  "timezone": "Africa/Lagos",
  "currency": "NGN",
  "trackingSince": "2026-09-20"
}
```

Render only the sections listed in sections .

9.3 Response envelope (every screen endpoint)

```
{
  "range": { "from": "2026-08-20", "to": "2026-09-18", "timezone": "Africa/Lagos", "interval": "day" },
  "compare": { "from": "2026-07-21", "to": "2026-08-19" },
  "generatedAt": "2026-09-18T10:42:07+01:00",
  "meta": {
    "trackingSince": "2026-09-20",
    "filters": ["source","device","state"],
    "definitions": { "visitors": "Distinct visitor ids with at least one event..." }
  },
  "data": { }
}
```

9.4 Building blocks inside data

```
// A KPI value
{ "key": "visitors", "label": "Visitors", "value": 1284, "previous": 1146, "change": 0.12,
  "format": "number",           // number | currency | percent | duration
  "goodDirection": "up",        // up | down | none
  "kind": null }                // cash | booked | estimate | liability | pending (money only)

// A time series
{ "key": "visitors", "points": [ { "t": "2026-09-01", "value": 42, "previous": 38 } ] }

// A breakdown row
{ "key": "google", "label": "Google (search)", "value": 312, "share": 0.243, "change": 0.08,
  "extra": { "conversion": 0.031 } }

// A funnel step
{ "key": "fee_paid", "label": "Fee paid", "count": 58, "fromPrevious": 0.62, "fromStart": 0.045,
  "medianSecondsFromPrevious": 5400 }
```

9.5 Endpoints

Method & path	data contains
GET /overview	kpis[6] , trend.visitors , trend.ideas , funnel[6] , attention[overdue[],transfers[],refunds[],stale[]} , topSources[5]

Method & path	data contains
GET /traffic	kpis[6] , newVsReturning{new,returning} , heatmap[7][24] (Mon-Sun x 0
GET /traffic/timeseries?metric=visitors sessions pageviews bounce_rate duration	series
GET /traffic/breakdown? dimension=source medium campaign referrer landing_page page device browser os country state&limit=50	rows[] , total
GET /engagement	kpis[] , interactions[] {event,target,count,visitors,ctr} , trackerSe scrollDepth{25,50,75,100} , downloads[]
GET /funnels/{application sales portal courses}?by=source device category	steps[] , overall , breakdown[] {key,label,steps[]}
GET /revenue 📈	cash{kpis[]} , booked{kpis[]} , series.netFees , byMethod[] . refunds{count,amount,medianDaysToRefund} , quotes{series[],acceptar page , pages)
GET /projects	kpis[] , byStage[] , timeInStage[] , deliveryVsTarget[] {code,title,da updateFrequency[] , table[]
GET /pipeline	kpis[] , series.submitted{online,walkIn} , by{category,platform,budg table[]
GET /clients	kpis[] , series.signIns , engagement{approvals,changeRequests,uploa optOuts{digest,promos} , byLocation[]
GET /courses	kpis[] , byCourse[] , bySource[] , counsellors[]
GET /team 📌	kpis[] , people[] , workload[]
GET /operations	messaging{series[],failureRate,topErrors[]} payments{attempts,successRate,abandoned,medianConfirmHours} , security{staffSignIns,failedSignIns,passwordResets,codesRequested, api{errors[],slowest[]}
GET /realtime	activeVisitors , perMinute[30] , topPages[] , topSources[] , devices[] {at,event,path,source,device}
GET /export?view=<screen>&table=<key>&format=csv xlsx	A file download (the same filters as the screen)
GET/POST /views , PATCH/DELETE /views/{id}	Saved views
GET/POST /schedules , PATCH/DELETE /schedules/{id}	Scheduled reports
POST /api/track (public)	Tracker ingest. {visitorId , sessionId , events[]} → 204 .

Caching. Responses carry Cache-Control: private, max-age=60 , except realtime (no-store). The client may reuse a response for 60 seconds when switching tabs.

10. Exports, saved views and scheduled reports

10.1 Exports

Format	How	Content
CSV	GET /export? ... &format=csv	One table. UTF-8 with BOM (so Excel shows ₦ correctly), , separator, ISO dates, raw numbers (no ₦ or % signs) plus a unit row.
Excel (.xlsx)	GET /export? ... &format=xlsx	One sheet per table on the screen, plus a "Read me" sheet with the range, filters, generated-at time and the definitions of the metrics included.
PDF	Built in the browser with the print stylesheet (@media print) or jsPDF	The whole screen: title, range, filters, tiles, charts (as SVG) and tables, with a footer giving the page number, generated-at time and "Confidential — Aptech". A4 portrait; wide tables switch to landscape.
PNG	Per chart, from the chart's ... menu	That chart only, with its title and range

Every export states the date range, filters and generation time. Nothing is exported without its definitions.

10.2 Saved views

- "Save view" stores the current screen and its query string under a name.
- A view is **Personal** by default. Admins can make it **Shared** with everyone who has analytics access.
- Saved views are listed in the sidebar. Opening one restores the exact URL.

10.3 Scheduled reports

- Created from any screen via ... → **Email this regularly**.
- **Frequency:** daily at 07:00, weekly on Monday at 07:00, or monthly on the 1st at 07:00 (Lagos time).
- **Recipients:** staff with analytics access only. External addresses are not allowed in v1.
- **Format:** PDF summary (default), CSV or Excel attached.
- **Dates:** each send uses a *rolling* range ("last 7 days"), not the dates that were on screen when it was set up.
- **Weekly business summary:** one is on by default for admins.

11. Design system and colour guide

The dashboard must look like part of AI Project Connect. Use the existing Tailwind v4 tokens from `app/globals.css` ; do not introduce new brand colours.

11.1 Brand tokens (existing)

Token	Hex	Use
navy-950	#06142a	Sign-in background, deepest surface
navy	#0b1f3a	Sidebar, top bar, primary text on light
navy-800	#112846	Sidebar hover
navy-700	#1a3561	Borders on navy
navy-600	#26467a	Secondary on navy
brand	#f26b22	Primary actions, active nav marker, the "Analytics" badge
brand-600	#e05a12	Primary hover
brand-700	#b9470a	Orange text on light (meets AA)
brand-soft	#fdebdd	Highlight backgrounds
teal	#14a38b	Positive accents
teal-700	#0d7a68	Positive text on light
teal-soft	#ddf2ee	Positive backgrounds
mist	#f3f5f9	Page background (light)
line	#e3e8f0	Card borders, hairlines
muted	#5b6b82	Secondary text, axis labels
blue-soft	#dfe7f2	Info backgrounds
danger	#b42318	Errors, destructive actions
danger-soft	#fbe3e1	Error backgrounds

Typography: Poppins (`font-display`) for headings, tile values and the hero numbers; Lato (`font-sans`) for everything else. Use `font-variant-numeric: tabular-nums` in tables and on axis ticks only.

Shape:

- Cards use `rounded-2xl` with a `1px line` border and a very soft shadow.
- Buttons are `rounded-xl` (44px tall for primary actions); pills are `rounded-full` .
- Spacing sits on a 4px grid: 16px gutters on mobile, 24-32px on desktop.

Light and dark themes: the app shell (sidebar, top bar, sign-in) is navy. Content areas are light by default (a white card on `mist`). A dark content theme is optional in v1 – if you build it, use the dark chart steps below and the surface `#0b1f3a` .

11.2 Chart palette – validated

The palette was checked with a colour-blindness validator:

- Protanopia and deuteranopia were simulated (Machado 2009).
- Separation is measured as Delta E in OKLab ×100: at least 8 for colour-blind vision and at least 15 for normal vision, for every pair of neighbouring colours.
- It was tested on the surfaces the dashboard actually uses: white `#ffffff` for light, and navy `#0b1f3a` for dark.

Assign the series slots in this order and never cycle them. The order is part of what makes the palette colour-blind safe.

Slot	Hue	Light (on <code>#ffffff</code>)	Dark (on <code>#0b1f3a</code>)
1	Brand orange	<code>#f26b22</code>	<code>#e05a12</code>
2	Teal	<code>#14a38b</code>	<code>#14a38b</code>
3	Blue	<code>#2a78d6</code>	<code>#3987e5</code>
4	Magenta	<code>#e87ba4</code>	<code>#d55181</code>
5	Yellow	<code>#eda100</code>	<code>#c98500</code>
6	Green	<code>#008300</code>	<code>#007a1e</code>
7	Violet	<code>#4a3aa7</code>	<code>#9085e9</code>
8	Red	<code>#e34948</code>	<code>#e66767</code>

Validator results:

Check	Light	Dark
All 8 inside the lightness band, chroma above the floor	PASS	PASS
Worst neighbouring pair, colour-blind vision	12.7 (teal–orange)	9.8 (green–yellow)
Worst neighbouring pair, normal vision	19.2	17.9
Contrast below 3:1 against the surface	magenta 2.7:1, yellow 2.2:1	green 2.99:1

Rules that come with the palette:

- 1. **Contrast relief.** Where a colour is under 3:1 (light: slots 4 and 5; dark: slot 6), charts using it must show **direct value labels** or **offer the table view**. Every chart already has a table view (see 13.3), which satisfies this.
- 2. **Scatter, bubble, map and small multiples:** use at most **3 series** (slots 1–3). Those three pass every-pair checks (colour-blind 12.7, normal 19.2). Beyond three, fold the rest into "Other" or split the chart.
- 3. **More than 8 series is not allowed.** Show the top 7 plus "Other" in grey #8a96a8.
- 4. **One series means one colour.** A single-series bar chart uses slot 1 for every bar. Never colour bars by their value.
- 5. **Colour follows the thing, not its rank.** "Paystack" is always the same slot on every screen, even when a filter hides other methods.
- 6. **Text is never series-coloured.** Values, labels and legends use `navy` / `muted`; a coloured swatch beside the text carries the identity.

Fixed series assignments (use these on every screen so people learn them):

Thing	Slot
Current period	1 (orange)
Comparison period	The same hue, dashed 1.5px, 60% opacity
Paystack · Bank transfer · At centre	1 · 2 · 3
Online · Walk-in	1 · 3
Desktop · Mobile · Tablet	3 · 1 · 2
Cash · Booked · Estimate	teal-700 #0d7a68 · navy #0b1f3a · muted #5b6b82 , used as tile label chips, not chart series

11.3 Sequential, ordinal and diverging

Job	Use	Colours
Sequential (heatmap: visits by hour, map shading)	One hue, light → dark	Blue ramp #cde2fb #9ec5f4 #6da7ec #3987e5 #2a78d6 #1c5cab #104281 #0d366b . Zero = <code>mist</code> #f3f5f9 .
Ordinal (funnel steps, ageing buckets)	Ordered steps of one hue	Blue steps 250 → 700: #86b6ef #5598e7 #2a78d6 #1c5cab #104281 #0d366b . The lightest step must still show against white (≥ 2:1), so never start lighter than #86b6ef .
Diverging (days early/late, change vs previous)	Two hues around a grey middle	Early / better = blue #2a78d6 ; late / worse = red #e34948 ; middle #e3e8f0 . Same number of steps each side.

11.4 Status colours (fixed, never themed)

Status always carries an **icon + label**; colour is never the only signal.

Status	Colour	Icon	Example
Good	#0ca30c (text #006300)	✓ check	Delivered on time
Warning	#fab219	! triangle	Transfer waiting for 2+ days
Serious	#ec835a	! circle	Project overdue
Critical	#d03b3b (text #b42318)	× octagon	Payment failures spiking

Deltas on tiles:

- **Good change:** teal-700 text on teal-soft.
- **Bad change:** danger text on danger-soft.
- **No change or neutral metric:** muted text.

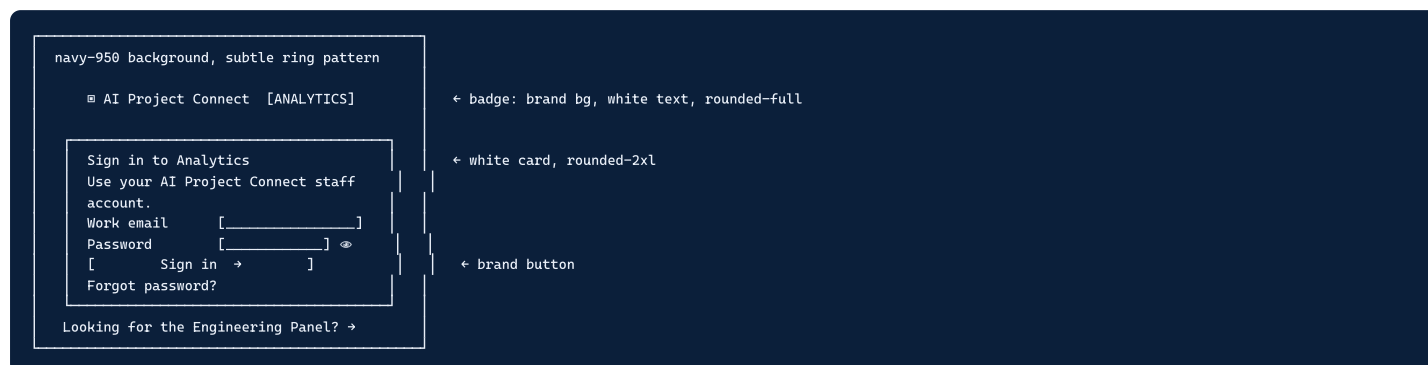
The arrow shows direction (▲/▼); the colour shows good or bad, from `goodDirection`.

11.5 Chart anatomy

- **Marks:**
 - Lines are 2px with a 4px rounded end. Markers are at least 8px and only on hover or the last point.
 - Bars have 4px rounded data-ends, anchored to a zero baseline, with a 2px gap between bars.
 - Stacked segments are separated by a 2px surface-coloured gap.
- **Axes and grid:** gridlines are a `line` #e3e8f0 hairline, horizontal only. Axis labels are `muted`, 12px. Keep ticks few (4–6). The y-axis always starts at 0 for bars.
- **Labels:** label selectively — the last point, max and min — never every point. Direct-label series when there are ≤ 4 of them; otherwise use a legend. Charts with 2 or more series always have a legend.
- **Never:**
 - Dual y-axes: two scales means two charts.
 - Pie charts with more than 4 slices, 3D effects or gradients.

- Rainbow scales.
- **Hover:**
 - Line and area charts get a crosshair and a tooltip listing every series at that point, including the comparison value and the delta.
 - Bars and heatmap cells get a per-mark tooltip.
 - Hit targets are larger than the mark (at least 24px).
- **Empty data:** show a neutral message inside the chart frame ("No visits in this range"), not an empty axis.

11.6 Sign-in page



12. Chart catalogue

Data	Chart	Why
A headline number	KPI tile (value, delta chip, sparkline of the range)	One number needs no axes
A metric over time	Line (area only for a single cumulative series)	Change over time
Two metrics over time with different scales	Two stacked small charts sharing the x-axis	Never a dual axis
Share of a whole with ≤ 4 parts	Horizontal 100% bar	Easier to compare than a pie
Ranking (sources, pages, courses)	Horizontal bars, sorted, top N + Other	Long labels stay readable
Funnel	Horizontal bars in ordinal blue, with conversion % between steps	Order shown by colour steps
Day $\times$ hour	Heatmap in sequential blue	Magnitude over two dimensions
Early / late vs target	Diverging bars around zero	Sign carries meaning
Nigeria by state	Sorted bar chart. A choropleth map is optional.	Bars are more accurate than area
Money by period	Columns	Discrete periods
Detail	Table with sticky header, sorting, CSV/Excel	Always available

13. States, performance and accessibility

13.1 States

State	Show
Loading	A skeleton with the same shape as the content: tiles, chart frame, table rows. Never a bare spinner on the whole page. Never show <code>0</code> while loading.
Empty	A plain sentence of what's missing and why ("Tracking started on 20 Sep — pick a later range").
Error	Inline in the card: the server's message plus a Try again button. Other cards keep working.
Partial	If a comparison fails but the main data loads, show the data and "Comparison unavailable".
Stale	If the data is older than 10 minutes (the tab was left open), show "Updated 14 min ago · Refresh".

13.2 Performance budgets

Budget	Target
Overview first paint (warm cache)	< 1.5 s on a mid-range Android over 4G
Any screen request (p95)	< 800 ms for ≤ 90-day ranges; < 2 s for 2-year ranges
Dashboard JavaScript	< 250 KB gzipped beyond the shared app bundle. Lazy-load the charting library per screen.
Tracker script on the public site	< 3 KB gzipped, loaded <code>defer</code> , and it never blocks rendering

Suggested chart library: a light SVG library, e.g. `visx` or `Recharts` — your choice, but it must render SVG (for the PDF export) and support the anatomy rules above.

13.3 Accessibility (WCAG 2.2 AA)

- **Table view:** every chart has one, via a "Table" toggle, which is also the screen-reader path. SVG charts carry `role="img"` and an `aria-label` summary ("Visitors rose 12% to 1,284 over the last 30 days").
- **Keyboard:** full keyboard access — the date picker, tabs, table sorting, chart tooltips (arrow keys move between points) and menus.
- **Contrast:** text contrast is at least 4.5:1 (3:1 for large numbers). Colour is never the only signal (see 11.2 and 11.4).
- **Motion:** respect `prefers-reduced-motion`. No count-up animations then.
- **Language:** plain, sentence case, no jargon in labels. Technical terms live in the definitions drawer.

14. Privacy and security

- **Nigeria Data Protection Act 2023:**
 - Analytics is aggregate and pseudonymous.
 - Show a short cookie notice with an opt-out on the public site; honour Do Not Track and Global Privacy Control.
 - Record the notice in the privacy policy.
 - **No personal data in tracking.** Private link parameters are stripped (7.2).
 - **IP addresses:** used only for rate limiting and coarse location (country and state). They are never stored in analytics tables.
 - **Access control:**
 - Every endpoint checks the session and `canViewAnalytics` ; Revenue and Team also check `role = admin` .
 - The dashboard must not rely on hiding things in the UI.
 - **Exports:** scheduled reports go only to staff with access. Exported files state "Confidential".
 - **The audit log records** analytics sign-ins, exports and schedule changes (who, what, when).
 - **Retention:** raw events 13 months, request logs 30 days, rollups kept.
 - **No third-party scripts** on the dashboard or the tracker.
-

15. Acceptance criteria

The dashboard is done when:

1. Sign-in:

- `/analytics` shows the branded sign-in page.
- Staff credentials work.
- A user without `canViewAnalytics` sees the no-access message and nothing else.
- A non-admin never sees Revenue or Team, and the API refuses them.

2. **Screens:** every screen in section 4 renders with real API data, loading, empty and error states, and works from 360px to 1920px wide without horizontal page scroll.

3. **Numbers:** every tile matches the API exactly. No number is recomputed in the browser, except formatting.

4. **Compare:** turning compare on shows deltas on the tiles and dashed comparison lines; good/bad colour follows `goodDirection`.

5. **URLs:** every filter, range and view survives a page reload and a copy-pasted URL.

6. **Exports:** CSV and Excel downloads contain the same rows as the screen, with the range, filters and definitions; the PDF export of each screen is legible in A4.

7. **Saved views:** create, open, rename, delete and share (admin).

8. **Scheduled reports:** create, edit, pause and delete; a test send arrives with the right range.

9. **Colours:** charts use only the palette in 11.2–11.4 in slot order; the table view is available for every chart.

10. **Accessibility:** keyboard-only walkthrough of every screen passes; axe shows no serious or critical issues.

11. **Realtime:** updates every 15 s while visible and stops when the tab is hidden.

12. **Performance:** meets the budgets in 13.2 on the staging server.

16. Delivery plan

Phase	Owner	Scope
1. Back end	Core team	New tables, <code>POST /api/track</code> , all <code>GET /api/analytics/*</code> endpoints, rollups, exports, saved views, schedules, and tests. Plus the tracker script and <code>data-track</code> attributes on the public site and portal.
2. Shell and Overview	Dashboard developer	Sign-in, layout, navigation, global controls, URL state, Overview
3. Core screens	Dashboard developer	Traffic, Engagement, Funnels, Revenue, Projects, Pipeline
4. Remaining screens	Dashboard developer	Clients, Courses, Team, Operations, Realtime
5. Output	Dashboard developer	Exports, saved views, scheduled reports UI, print stylesheet
6. QA	Both	Acceptance criteria, accessibility pass, performance on staging

Working agreement:

- The API contract in section 9 is the interface. If a screen needs a number that isn't in a response, ask for it rather than deriving it in the browser.
- Until phase 1 ships, build against the JSON examples in section 9.4.
- Use the demo database (`php bin/setup.php --demo`) for realistic business data, plus the tracker seed (`--demo-analytics` , added in phase 1) for traffic.

17. Appendix

17.1 Glossary

Term	Meaning
Commitment fee	The one-off ₦2,000 (configurable) fee a client pays before an idea is submitted; refunded in full if the idea is declined
Draft	An application saved but not yet submitted
Walk-in	An application started by staff at an Aptech centre for a visitor
Booked value	Money agreed (accepted quotes, approved changes) but not collected through the platform
Stale project	An active project with no client-visible update for 5+ days
CTA	Call to action — a button meant to move someone forward
MRR	Monthly recurring revenue

17.2 Naming conventions

- Events: `snake_case` nouns (`page_view` , `cta_click`), with the variation in `name` / `props` .
- CTA ids: `<place>_<action>` (`hero_submit_idea`).
- Metric keys in the API: `camelCase` (`netFeeRevenue` , `onTimeRate`).

17.3 Related documents

- `README.md` — how the project runs and deploys
- `backend/README.md` — the full API reference for the existing platform
- `app/globals.css` — the design tokens

18. Build notes (v1.1)

The back end in section 9 is built, tested (132 automated checks) and running on the demo database. The API follows this spec, with the clarifications below. Build against the real responses; these notes explain the fields you will see.

18.1 Extra response fields

- **Filters:** every screen's `meta` also has `appliedFilters` (the filters the server actually used) and `includeInternal` (whether staff traffic is counted).
- **Money on shared screens:** it is admin-only. On Overview and Courses, a money tile comes back as `{ "value": null, "restricted": true }` for non-admins, and amounts in the Overview attention list are `null`. Render these as "Admin only", never as ₦0.
- **Attention rows** carry the record ids and `link: "/engineering"`. The panel has no deep links yet, so link to the panel.
- **Funnel notes:** a breakdown that doesn't apply (see 18.2) returns an empty `breakdown` plus `notes: ["..."]`. Show the note instead of an empty table.
- **Excel exports:** the `X-Export-Format` response header says `xlsx` or, on a server without the zip extension, `xls`. Name the downloaded file accordingly.

18.2 How some numbers are counted

Topic	What the API does
Visitors across a range	Exact distinct visitors for any range and filter, read from the session table
Long ranges	Days before yesterday come from nightly rollups; yesterday and today from raw events. Both paths give identical totals (tested).
Overview filters	The Overview takes no filters; it mixes traffic and business figures.
Application and portal funnels	Counted in tracked visitors. <code>by=category</code> is not possible for these two.
Sales funnel	"Reviewed" has no timestamp, so its median time is <code>null</code> . Ideas marked "quote sent" without a quote record count as sent but are left out of the medians.
Courses funnel	"Viewed" and "clicked" count visitors; "enquiry", "contacted" and "enrolled" count course leads.
Time to first contact / enrolled	From the new <code>leads接触ed_at</code> and <code>leads.enrolled_at</code> columns (back-filled for existing leads)
Security figures	Staff sign-ins, failed sign-ins and rate-limit hits come from the request log, so they cover the last 30 days only.
Location	<code>country</code> comes from Cloudflare's <code>CF-IPCountry</code> header when present. <code>state</code> is always <code>null</code> — shared hosting has no GeoIP. Hide the State breakdown until one is added.

18.3 Tracking

- **Private links:** paths keep only `utm_*`, `view`, `funnel` and `payment` in the query string. Everything else is removed on the server as well as in the browser.
- **CTA ids:** any id matching `^[a-z0-9_]{1,60}$` is accepted, so a new `data-track` id works without a back-end change.
- **Filtered requests:** requests dropped for Do Not Track, Global Privacy Control or bots still get `204`.
- **Retention:** raw events are kept for 13 months. Sessions and visitors are kept for 25 months, so exact visitor counts cover the full 2-year range.

18.4 Scheduled reports

A PDF schedule sends an HTML summary email with a link to the screen, because the PDF itself is made in the browser. CSV and Excel schedules attach the file.

18.5 Running it locally

```
php bin/setup.php --demo --demo-analytics # fresh demo data plus ~90 days of traffic
php bin/analytics-rollup.php # nightly job; safe to run again
php tests/analytics.php http://127.0.0.1:8088
```

Sign in at `/analytics` with `admin@aptechdevteam.com` / `Aptechdev123` once your screens exist. Until then, call the endpoints directly with that session.

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